

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 9

DUE: WEDNESDAY, APRIL 8, 2026, **BEFORE 23:59PM**

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. Let $\tilde{H}_p(X)$ be reduced homology of X , defined as in HW7. Let us write $i : \tilde{H}_p(X) \rightarrow H_p(X)$ for the homomorphism defined there, induced by the chain map $\tilde{C}_*(X) \rightarrow C_*(X)$.

- (a) Let $j : H_p(X) \rightarrow H_p(X, A)$ be the homomorphism in the long exact sequence of the pair (X, A) . Prove that for $x \in X$ and $A = \{x\}$, the composition

$$\tilde{H}_p(X) \xrightarrow{i} H_p(X) \xrightarrow{j} H_p(X, \{x\})$$

is an isomorphism for all p .

- (b) Deduce an isomorphism

$$\tilde{H}_p(X/A) \cong H_p(X, A)$$

for all p , whenever (X, A) is a good pair.

2. Let $\{*\}$ be a one-point space, and as usual write $I = [0, 1]$. The *cone* of a topological space X is defined as

$$CX = (I \times X)/(\{1\} \times X),$$

and write $*$ $\in CX$ for the basepoint, so that $\{*\} = (\{1\} \times X)/(\{1\} \times X)$.

- (a) Prove that the inclusion

$$\{*\} \xrightarrow{j} CX$$

is a homotopy equivalence for any X .

- (b) Let

$$i : X \rightarrow CX$$

be defined by $i(X) = [(0, x)]$, the equivalence class of $(0, x) \in I \times X$. Prove that the induced map $X \rightarrow i(X)$ is a homeomorphism,

when $i(X) \subset CX$ is given the subspace topology. Then prove that $(CX, i(X))$ is a good pair.

3. This question is a continuation of the previous.

- (a) Use the long exact sequence of the pair $(CX, i(X))$ to construct an isomorphism

$$\tilde{H}_p(X) \rightarrow H_{p+1}(CX, i(X))$$

for any X and any $p \in \mathbb{Z}$.

- (b) Let $SX = CX/i(X)$ have the quotient topology. Deduce an isomorphism

$$\tilde{H}_p(X) \cong \tilde{H}_{p+1}(SX)$$

for all p and any space X .

Remark (not part of the question): the space SX is usually called the suspension of X . This construction can be promoted to a functor $S : \mathbf{Top} \rightarrow \mathbf{Top}$, and satisfies $S(S^n) \approx S^{n+1}$ for all $n \geq -1$.

4. Let A be a topological space and $e : \partial D^n \rightarrow A$ a continuous map. In class, we have explained how to attach a cell to A along e .

- (a) Prove that if A is compact, then $A \cup_e D^n$ is also compact. Then prove that if A is Hausdorff, then $A \cup_e D^n$ is also Hausdorff.
- (b) Let X be a compact Hausdorff space and $f : D^n \rightarrow X$ a continuous map, and assume that $f(D^n \setminus \partial D^n) \subset X$ is open and that $f(v) \neq f(v')$ whenever $v \neq v'$ and $v \in D^n \setminus \partial D^n$. Construct a homeomorphism

$$A \cup_e D^n \rightarrow X,$$

where $A = X \setminus f(D^n \setminus \partial D^n)$ and $e = f|_{\partial D^n}$.

5. In this problem we write $S^{2n+1} = \{v \in \mathbb{C}^{n+1} \mid |v|^2 = 1\}$, where the length of a vector $v = (z_0, \dots, z_n)$ is defined as $|v|^2 = \sum_{i=0}^n |z_i|^2$. Define an equivalence relation $\sim$ on S^{2n+1} as $v \sim v'$ if and only if $v' = uv$ for some $u \in \mathbb{C}$ with $|u| = 1$. The quotient space $S^{2n+1}/\sim$ is usually denoted $\mathbb{C}P^n$, and comes with a quotient map $q : S^{2n+1} \rightarrow \mathbb{C}P^n$. Furthermore, let $f : D^{2n} \rightarrow \mathbb{C}P^n$ be defined as

$$f(v) = q(\sqrt{1-|v|^2}, v).$$

In this problem, it may be used without proof that the assumptions in the previous question hold for this map (with $X = \mathbb{C}P^n$), i.e., that $f(D^{2n} \setminus \partial D^{2n}) \subset X$ is open and $f(v) \neq f(v')$ whenever $v \neq v'$ and $v \in D^{2n} \setminus \partial D^{2n}$, and that $\mathbb{C}P^n$ is Hausdorff.

- (a) Explain why $A = \mathbb{C}P^n \setminus f(D^n \setminus \partial D^n)$ is homeomorphic to $\mathbb{C}P^{n-1}$ and deduce a homeomorphism

$$\mathbb{C}P^n \approx \mathbb{C}P^{n-1} \cup_e D^{2n}.$$

- (b) Deduce that $H_p(\mathbb{C}P^n) \cong H_p(\mathbb{C}P^{n-1})$ for $p \notin \{2n-1, 2n\}$.
- (c) It can be shown that $H_p(\mathbb{C}P^{n-1}) \cong 0$ for $p \geq 2n-1$. Use this to deduce a formula for $H_p(\mathbb{C}P^n)$ for all $p \geq 2n-1$.

Remark (not part of the question): Since we know the homology of $\mathbb{C}P^0 = \{\}$ in all degrees, this question can be used to establish a formula for $H_p(\mathbb{C}P^n)$ for all p and n , by induction on n .*